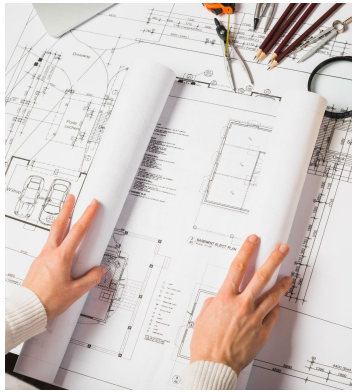
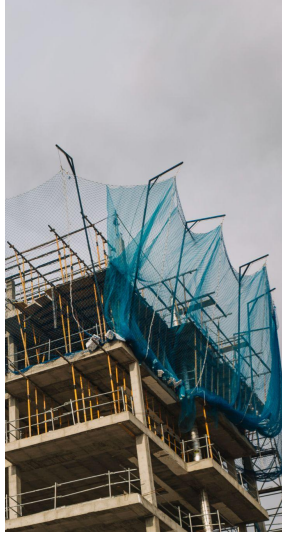
Visualising Software Architecture

Software Engineering Course – 4
2024-2025





Agenda

01

Recap Software Architecture

02

Why visualisation

03

UML

04

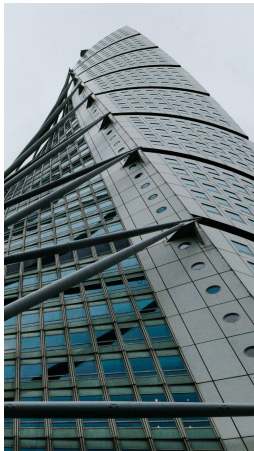
C4 Model

05

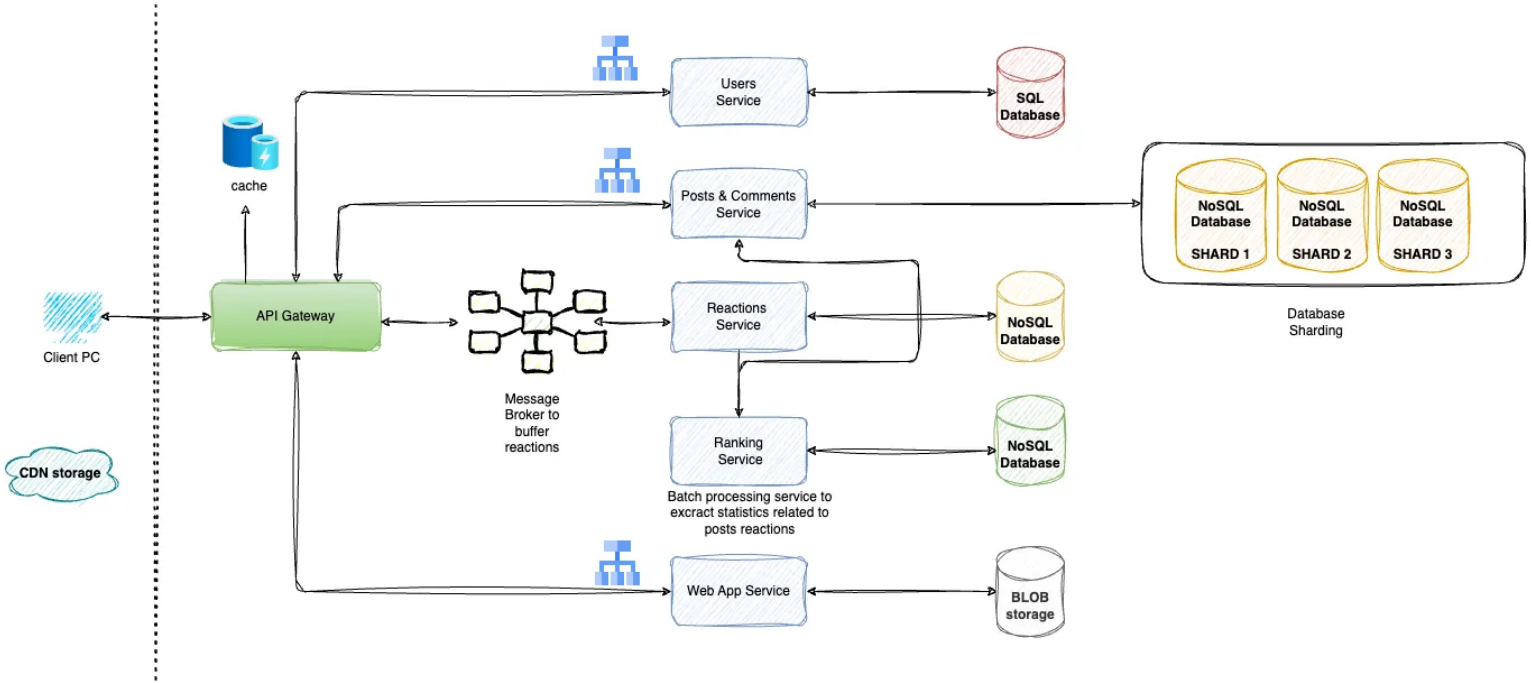
Context and Container Diagrams

06

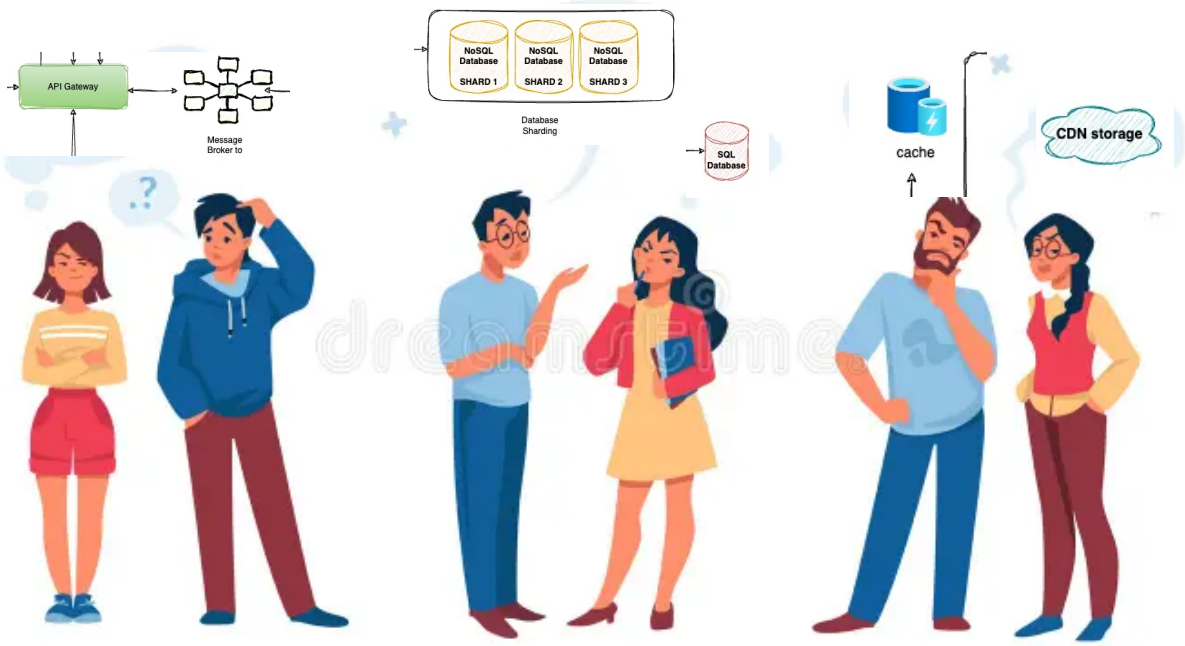
Tools

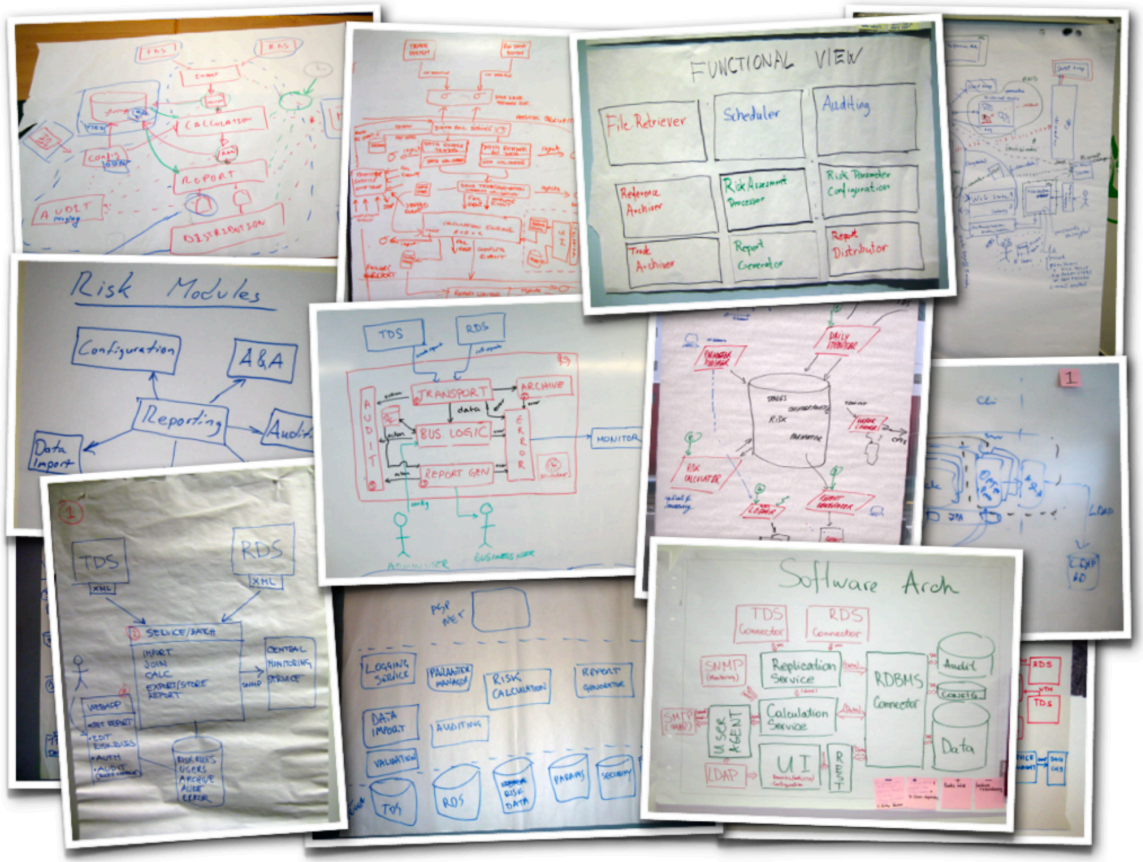


Software Architecture



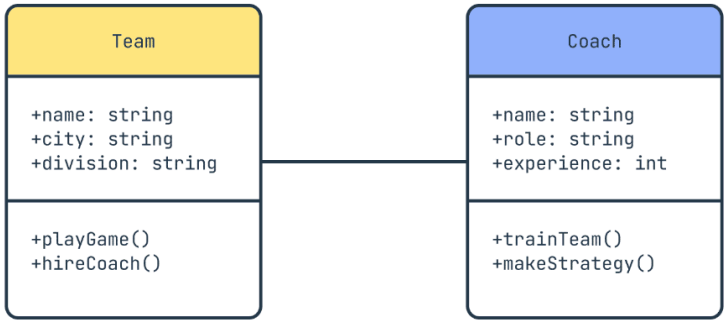
The failure to communicate

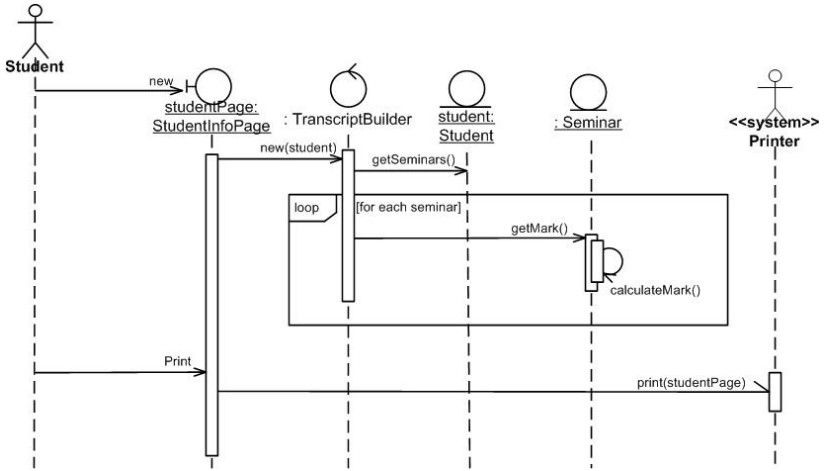


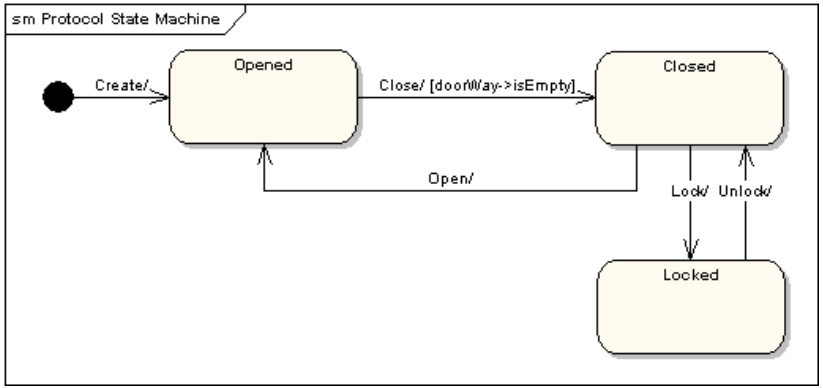


A selection of typical “boxes and lines” diagrams

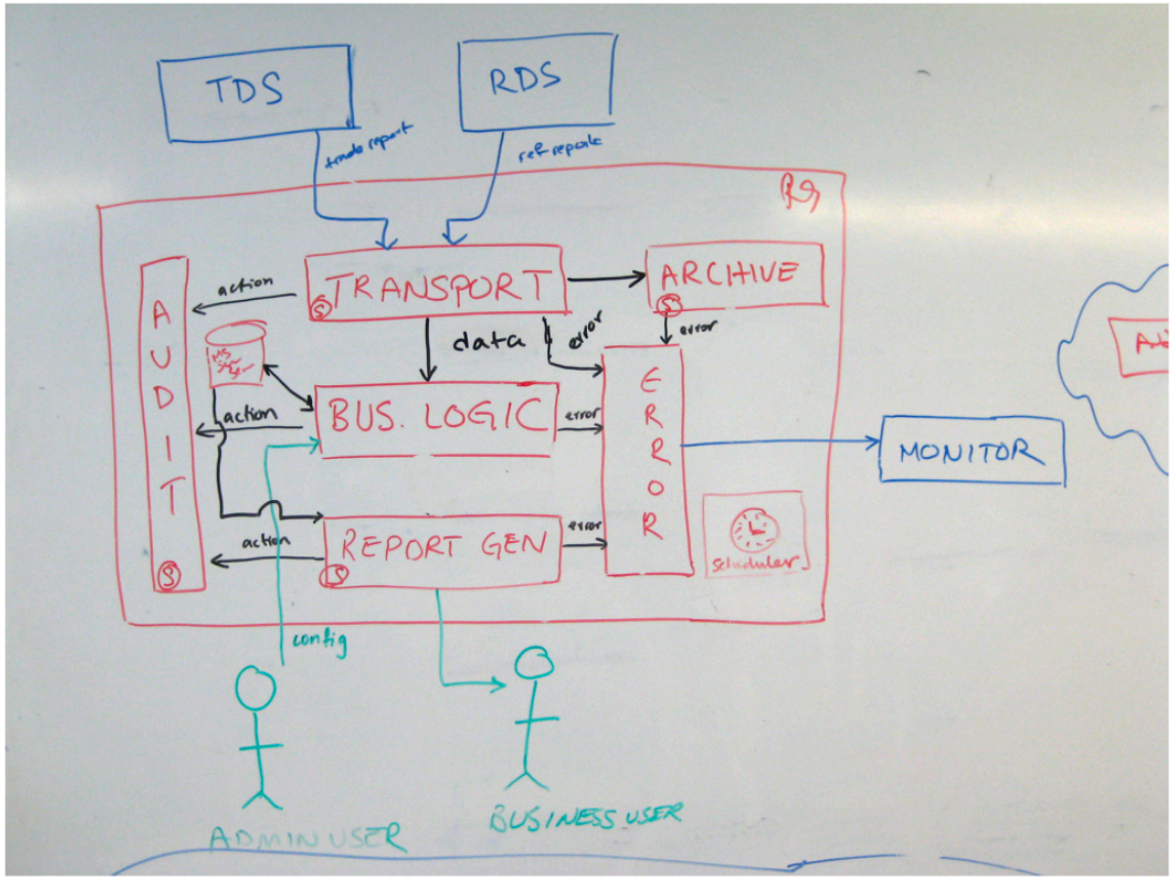
UML


Team
+name: string
+city: string
+division: string
+playGame()
+hireCoach()
Coach
+name: string
+role: string
+experience: int
+trainTeam()
+makeStrategy()

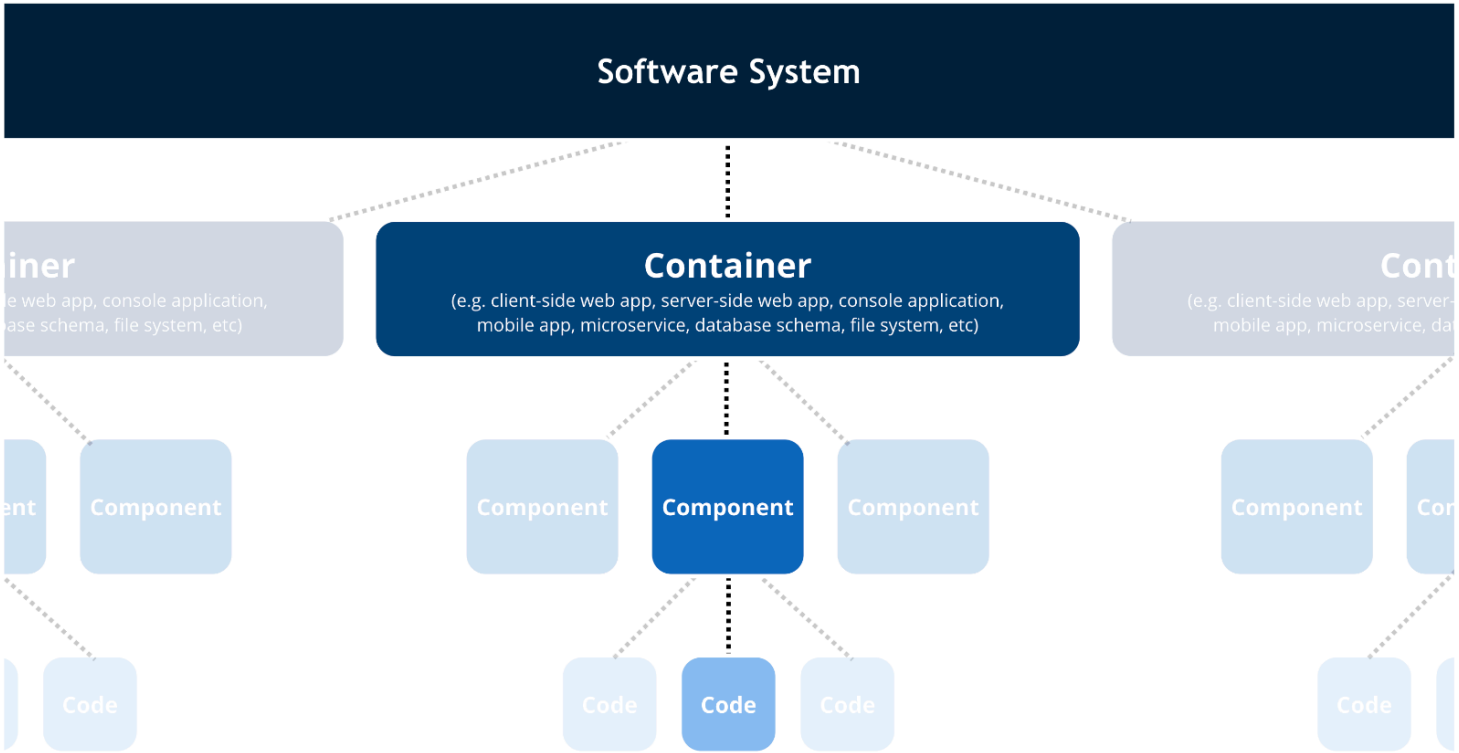

Student
new
studentPage:
StudentInfoPage
: TranscriptBuilder
student:
Student
: Seminar
<<system>>
Printer
new(student)
getSeminars()
loop
[for each seminar]
getMark()
calculateMark()
Print
print(studentPage)


sm Protocol State Machine
Create/
Opened
Close/ [doorWay->isEmpty]
Closed
Open/
Lock/
Unlock/
Locked

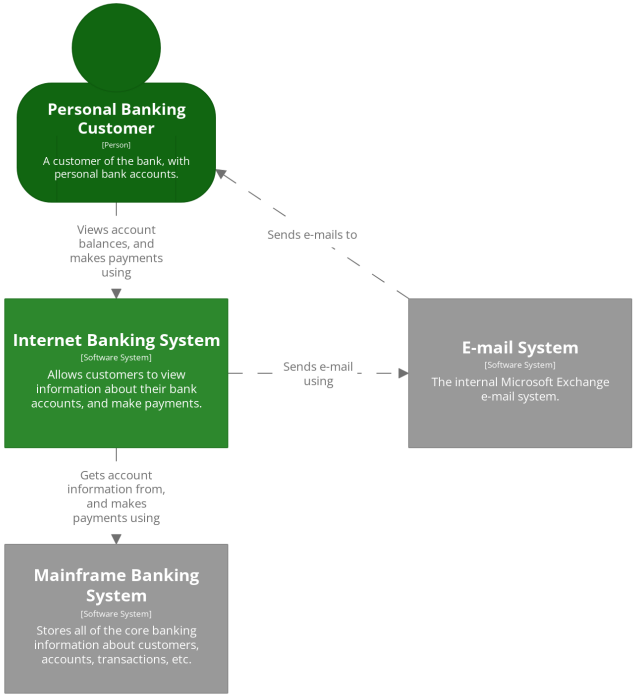
Attempts were made



Common vocabulary



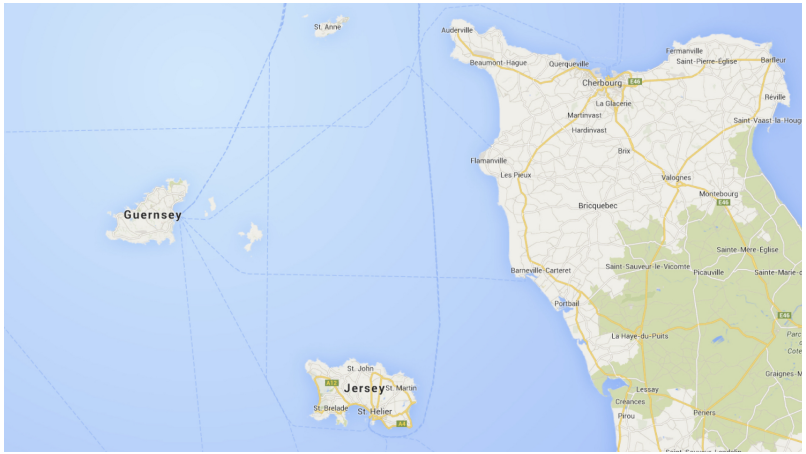
Maps of your code - Context Diagram

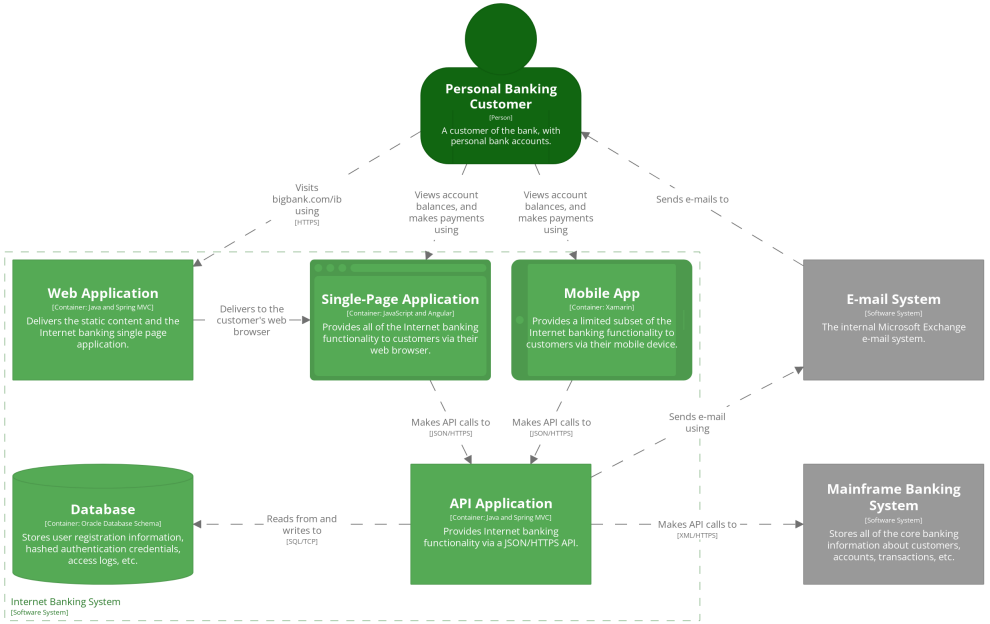


[System Context] Internet Banking System

The system context diagram for the Internet Banking System - diagram created with Structurizr.
Wednesday, March 22, 2023 at 8:16 AM Coordinated Universal Time

Maps of your code - Container Diagram

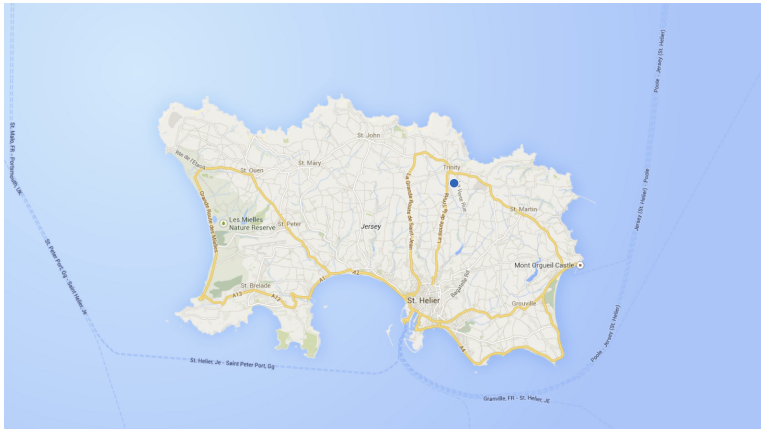


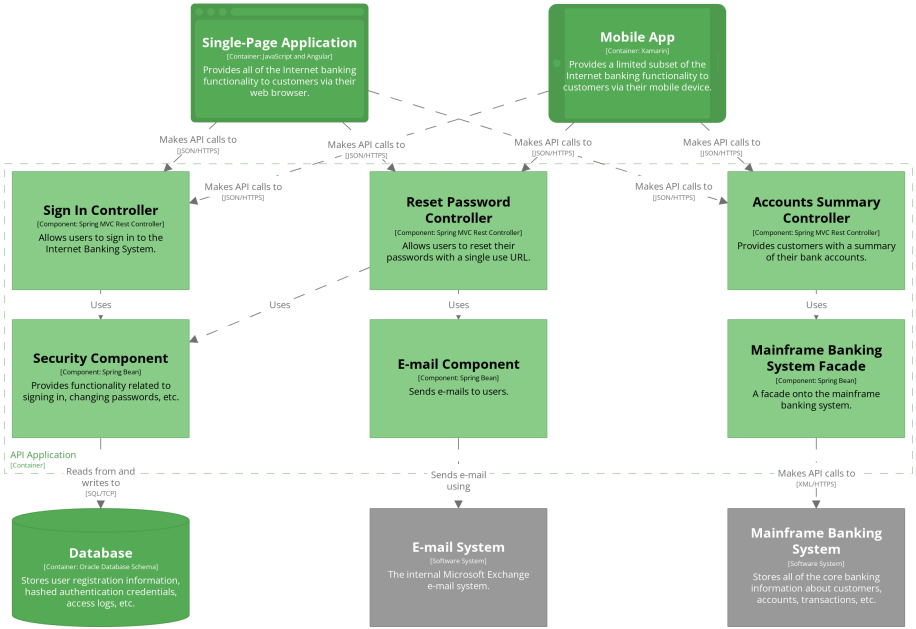


[Container] Internet Banking System

The container diagram for the Internet Banking System - diagram created with Structurizr.
Wednesday, March 22, 2023 at 8:16 AM Coordinated Universal Time

Maps of your code - Component Diagram



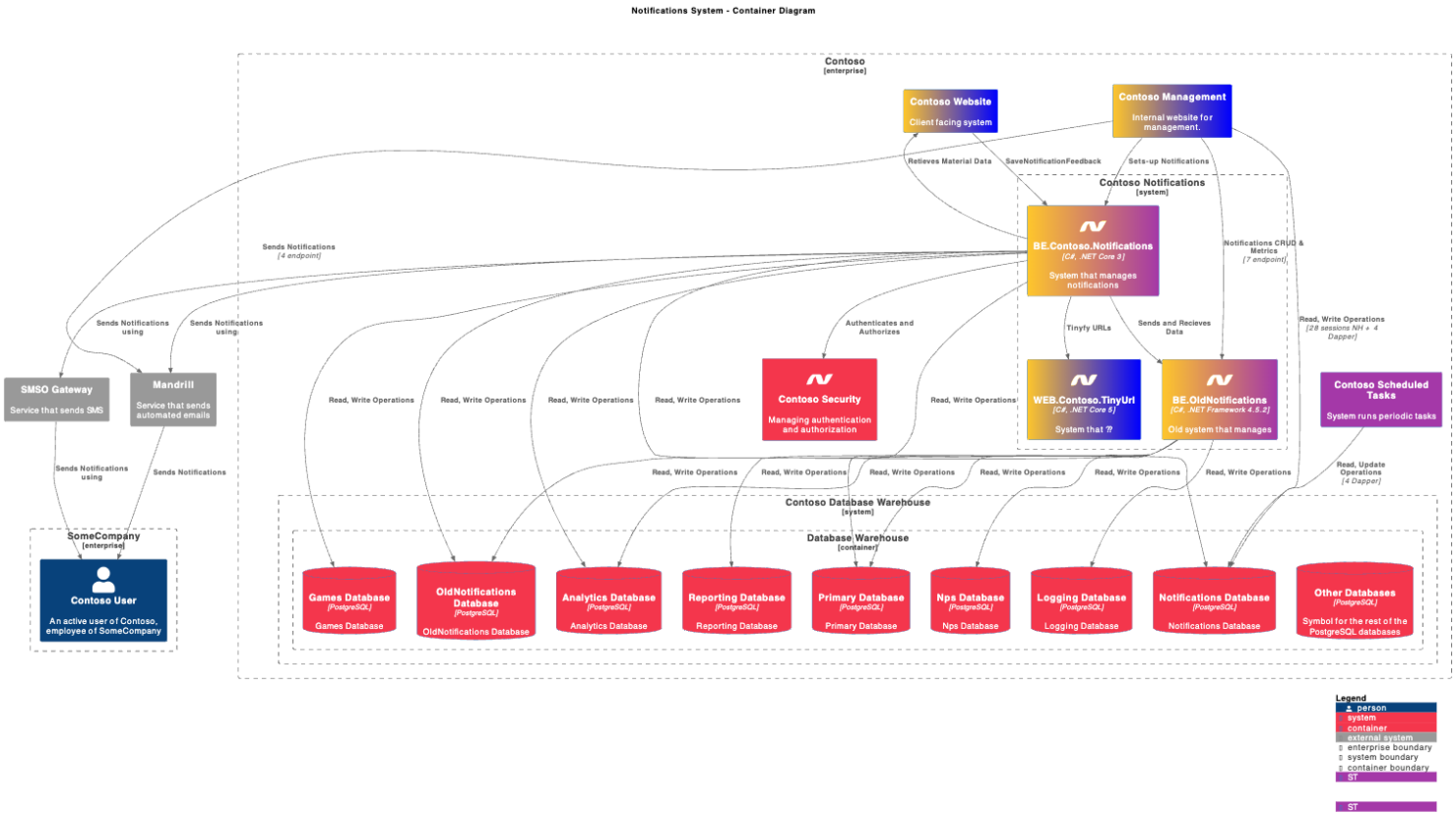


[Component] Internet Banking System - API Application

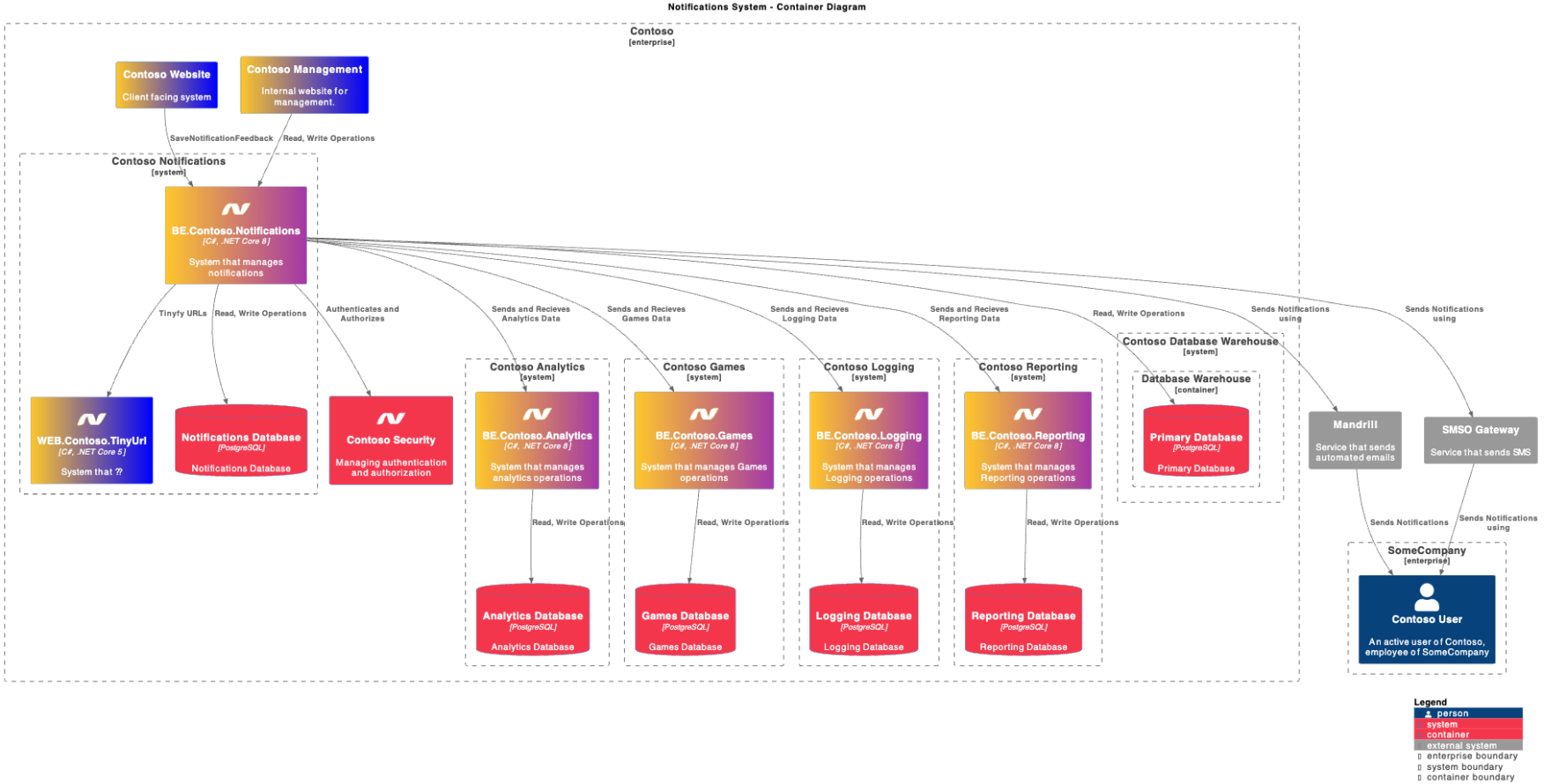
The component diagram for the API Application - diagram created with Structurizr.

Wednesday, March 22, 2023 at 8:16 AM Coordinated Universal Time

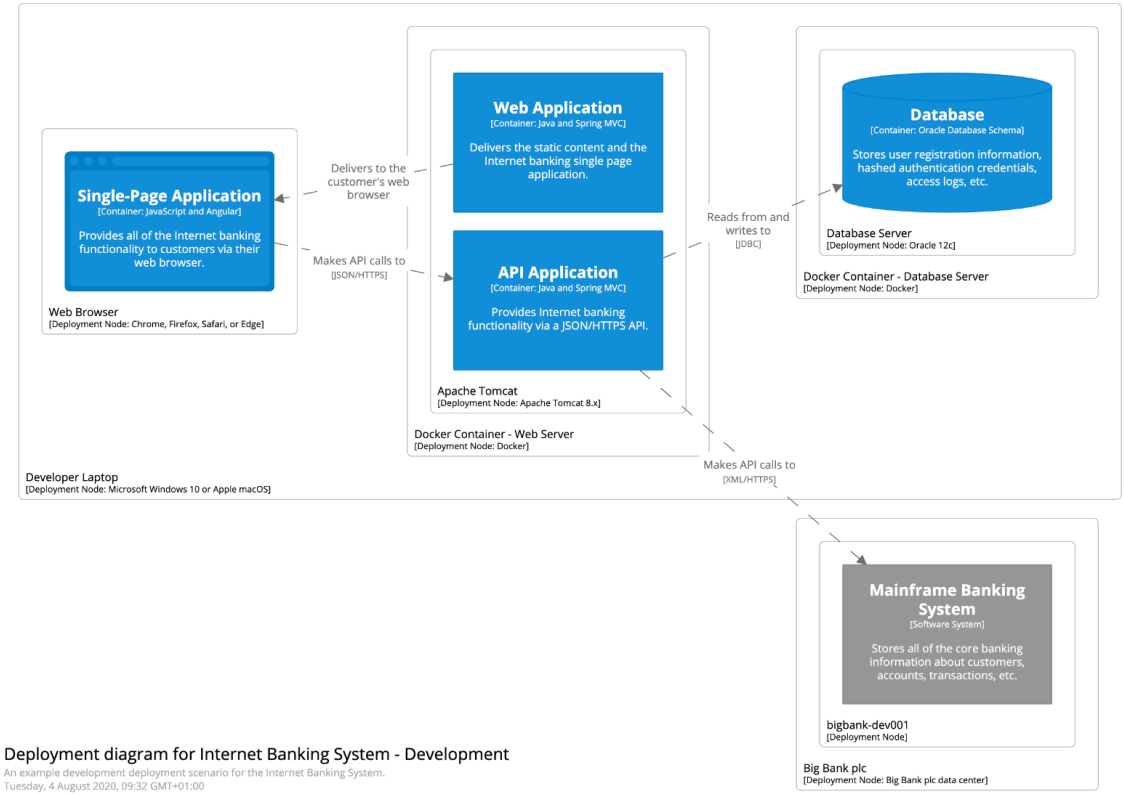
Real-Case Example



Real-Case Example



Deployment Diagram



Deployment diagram for Internet Banking System - Development

An example development deployment scenario for the Internet Banking System.
Tuesday, 4 August 2020, 09:32 GMT+01:00

Kahoot!

Demo

Next time on …

3.11 - Deadline for Interim Deliverable

Course 6

Please read from *Engineering Software Products (Sommerville)*:

- Chapter 6 – Microservices (p.160-164)
- Chapter 6.3 - RESTful services (p.183- 189)